

The Indian numbering system is used in India, Pakistan, Nepal, Sri Lanka, and Bangladesh to express large numbers, which differs from the International System of Units. Commonly used quantities include lakh (one hundred thousand, 10⁵) and crore (ten million, 10⁷) – written as 1,00,000 and 1,00,00,000 respectively in some locales. For example: 150,000 rupees is "1.5 lakh rupees" which can be written as "1,50,000 rupees", and 30,000,000 (thirty million) rupees is referred to as "3 crore rupees" which can be written as "3,00,00,000 rupees".

There are names for numbers larger than crore, but they are less commonly used. These include arab (100 crore, 10⁹), kharab (100 arab, 10¹¹), nil or sometimes transliterated as neel (100 kharab, 10¹³), padma (100 nil, 10¹⁵), shankh (100 padma, 10¹⁷), and mahashankh (100 shankh, 10¹⁹). In common parlance (though inconsistent), the lakh and crore terminology repeats for larger numbers. Thus lakh crore is 10¹².

In the ancient Indian system, still in use in regional languages of India, there are single-word names for up to 10⁶² (see § Historic numbering systems). In the Indian system, now prevalent in the northern parts, the next powers of ten are one lakh, ten lakh, one crore, ten crore, one arab (or one hundred crore), and so on.

Multiples

The Indian system is decimal (base-10), same as in the International System of Units, and the first five orders of magnitude are named in a similar way: one (10⁰), ten (10¹), one hundred (10²), one thousand (10³), and ten thousand (10⁴). For higher powers of ten, naming diverges. The Indian system uses names for every second power of ten: lakh (10⁵), crore (10⁷), arab (10⁹), kharab (10¹¹), etc. In the long and short scales, there are names for every third power of ten. The short scale uses million (10⁶), billion (10⁹), trillion (10¹²), etc.

Decimal formatting

In the Indian system, digits of a large decimal are grouped differently than in the International

System of Units. Both systems indicate the first three digits to the left of the decimal point as a group. Thereafter, the Indian system groups by two digits. This aligns with the naming of quantities at multiples of 100.

Like English and other locales, the Indian system uses a period as the decimal separator and the comma for grouping, while others use a comma for decimal separator and a thin space or point to group digits.

Pronunciation in English

When speakers of indigenous Indian languages are speaking English, the pronunciations may be closer to their mother tongue (e.g. "lakh" and "crore" might be pronounced /lɑ:kʰ/, /kə.ro:ʃ/, respectively).

lakh /lɑ:kʰ/

crore /kə.ro:ʃ/ (or /kɹoʊ/ in American English)

arab /ə.rəb/

kharab /kʰə.rəb/

Names of numbers

The table below includes the spelling and pronunciation of numbers in various Indian languages along with corresponding short scale names.

Historic numbering systems

Numbering systems in Hindu epics

There are various systems of numeration found in various ancient epic literature of India (itihasas). The following table gives one such system used in the Valmiki Ramayana.

Other numbering systems

The denominations by which land was measured in the Kumaon Kingdom were based on arable lands and thus followed an approximate system with local variations. The most common of these was a vigesimal (base-20) numbering system with the main denomination called a bisi (see Hindustani number bīs), which corresponded to the land required to sow 20 nalis of seed. Consequently, its actual land measure varied based on the quality of the soil. This system became the established norm in Kumaon by 1891.

Usage in different languages

Below is a list of translations for the words lakh and crore in other languages spoken in the Indian subcontinent:

Formal written publications in English in India tend to use lakh/crore for Indian currency and International numbering for foreign currencies.

Current usage

The official usage of this system is limited to the nations of India, Pakistan and Bangladesh. It is universally employed within these countries, and is preferred to the International numbering system.

Sri Lanka and Nepal used this system in the past but have switched to the International numbering system in recent years. In the Maldives, the term lakh is widely used in official documents and local speech. However, the International System of Units is preferred for higher denominations (such as millions).

Most institutions and citizens in India use the Indian number system. The Reserve Bank of India was noted as a rare exception in 2015, whereas by 2024 the Indian system was used for

amounts in rupees and the International system for foreign currencies throughout the Reserve Bank's website.

See also

Scientific notation

Japanese notation

References

Notes

References